

Secure Reactive Routing Protocol using On-Demand Dynamic Mobile Networks

K. Thamizhmaran

Assistant Proffessor, Dept. of ECE, GCE, Bodinayakkanur, India

Corresponding Author

E-Mail Id: tamil10_happy@rediff.comaa

ABSTRACT

There square measure finding a route to destination, new route discovery is initiated. The frequent discoveries result in a lot of network congestion. To avoid this multipath routing protocols are planned to seek out multiple routes to destination and put on to alternate secondary path just in case of route broken and its offer higher routing performance and security. In this developed research paper attempt to compare simulation performance of two on demand protocols Secure Dynamic Source Routing (SDSR) and traditional Dynamic Source Routing (DSR) with different topology and malicious nodes, on-demand gateway discovery protocol where a mobile device of MANET gets connected to gateway. SDSR using the Intrusion Detection system (IDS) and trust based routing, the performance results are analysed by varying simulation time. Furthermore, all the above mentioned protocols are compared based on several important performance metrics which are packet delivery ratio, end-to-end delay and remaining energy through NS2.

Keywords: *SDSR, result, routing, protocols, destination*

INTRODUCTION

The recent trends in wireless communication network have changed the lives of human beings. The new wireless technologies create a potential for the next generation MANET and applications. Wireless technologies such as Bluetooth or the 802.11 standards enable mobile devices to establish a MANET by connecting dynamically through the wireless medium without any centralized structure. MANET is a network having dynamic topology that consists of mobile nodes without base Station or centralized control. All mobile nodes perform functioning of routers that search and maintain routes to other nodes in the network. Difficulty in ad-hoc network is that for communication with other nodes, a node must be in the transmission range of base station but sometimes a node moves and network fails. MANET has solved this problem as in MANET nodes follow

multi-hop pattern for communicating with other nodes. Routing is the process of moving information across internet work from source to destination by selecting best outgoing path that a packet has to take in internetwork. To perform this, a set of routing protocols needed that uses metrics to find optimal path for a packet to travel Routing protocols planning objectives are optimality, effortlessness, low above, heartiness, unwavering quality and adaptability. Course solicitation and Course answer messages are utilized to find and store the ways found from the source to objective. Subsequent to finding the ways, most limited way is chosen by the source node. Paths discovered by shortest path algorithm cause problems like congestion problems as the centre of network carry more traffic, this results in poor performance. The principal objectives of multipath directing conventions are to

keep up with solid correspondence, to diminish steering above by utilization of auxiliary ways, to guarantee load adjusting, to work on nature of administration, to stay away from the extra course revelation above. The major focus of the routing is the performance and the efficiency of the protocol in the presence of a dynamic network environment. The routing protocol has to overcome the security pitfalls to utilize the potentials of the MANET. A secure routing is challenging due to the security vulnerabilities present in the active network.[1-4]

BACKGROUND WORK

This work evolves the previous work that has been undergone in this field. An execution investigation of AODV steering convention was finished by Das, *et al.* (2000). Improvement and examination of multipath steering convention AOMDV in view of CMMBCR was finished by Yin-jun Yang (2011). A hub disjoint multipath directing technique in light of AODV convention for MANET was finished by Lal, *et al.* (2012). Expanded throughput for load based direct mindful steering in MANETs with reusable ways was finished by Ayyasamy (2012). Particular based interruption discovery for automated airplane frameworks was finished by Mitchell (2012). An overview of assaults on MANET steering conventions was finished by Tayal (2013).

Audit on MANET steering conventions and difficulties were finished by Habib (2013). A study of interruption location in remote organization applications was finished by Mitchell (2014). Conduct rule detail based interruption location for wellbeing basic clinical digital actual frameworks was finished by Mitchell (2015). Confided in secure impromptu on-request multipath distance vector steering in MANET was examination by Abrar Omar Alkhamisi (2016).[5-10]

EXISTING METHOD

DSR (Dynamic Source Routing)

In DSR, when a course is required from source to objective, then, at that point, source begins a course disclosure process by flooding a RREQ for objective. With the assistance of arrangement numbers, RREQs are remarkably distinguished so that copy RREQs can be recognized and disposed of. At the point when a non-copy RREQ is gotten then middle of the road hub records past bounce and quest for a new course passage to the objective in directing table. In the event that new course is available, the hub sends a RREP to the source yet in the event that new course is absent, it rebroadcasts the RREQ. The steering data is refreshed by a hub provided that a RREP contains either a bigger objective succession number than past one or a course with less bounce count found. DSR routing process with the help of IDS and trust-based routing, the attack identification and isolation in DSR are carried out in two phases of routing such as route discovery phase and data forwarding phase. The trust obtained from IDS is applied in the routing decision-making about propagating RREQ packet of the source and selecting the trust based router for data forwarding. Trust based route discovery process initially, each node assigns the trust value as 'one' to its neighboring nodes. According to the routing activities of these nodes, the IDS measure the original trust value and inform the network layer. The route discovery process is carried out based on the trust value to isolate the flooding attacker activity. Prior to rebroadcasting the received RREQ packet to the neighbors, every node checks for the trust value of the source that has broadcasted the RREQ packet. If the trust value is lesser than the threshold, then the RREQ packet from the corresponding source is dropped to block the flooding activity of the attacker. For instance, in the trust value of the source node is maintained by its neighboring

nodes such as A, B, and C. The trust value of the source node is different in various neighboring nodes due to the network collisions. The trust value of the source node is very less, so these nodes do not forward the RREQ packet of the source node into the network. Trust based data forwarding process data forwarding process is carried out based on the trust value to isolate the activity of black hole and the grayhole attacker. Prior to forwarding the data packet to the router, every node checks for the trust value of the router. The trust value of the router indicates the reliability of data delivery through it. If the trust value is low, the current data transmission through the malicious router is blocked. Subsequently, the trusted router from the routing table is accessed to resume data transmission through it. For instance, the trust value of the node B is higher than node A, and so the source node selects router B for further data forwarding. Thus, the IDS and the trust based DSR protocol improve the routing performance and security in MANET.[11,12]

PROPOSED METHOD

SDSR (Secure Dynamic Source Routing)

In this proposed work TS-DSR, when a route is needed from source to destination, then source starts a route discovery process by flooding a RREQ for destination. With the assistance of arrangement numbers, RREQs are remarkably distinguished so that copy RREQs can be recognized and disposed of. At the point when a non-copy RREQ is gotten then middle of the road hub records past bounce and quest for a new course passage to the objective in directing table. In the event that new course is available, the hub sends a RREP to the source yet in the event that new course is absent, it rebroadcasts the RREQ. The steering data is refreshed by a hub provided that a RREP contains either a bigger objective

succession number than past one or a course with less bounce count found. SDSR routing process with the help of IDS and trust-based routing, the attack identification and isolation in SDSR are carried out in two phases of routing such as route discovery phase and data forwarding phase. Trust Based Route Discovery Process Initially, each node assigns the trust value as 'one' to its neighboring nodes. Prior to rebroadcasting the received RREQ packet to the neighbors, every node checks for the trust value of the source that has broadcasted the RREQ packet. If the trust value is lesser than the threshold, then the RREQ packet from the corresponding source is dropped to block the flooding activity of the attacker. For instance, in the trust value of the source node is maintained by its neighboring nodes such as A, B, and C.

The trust value of the source node is different in various neighboring nodes due to the network collisions. The trust value of the source node is very less, so these nodes do not forward the RREQ packet of the source node into the network. For instance, the trust value of the node B is higher than node A, and so the source node selects router B for further data forwarding. Thus, the IDS and the trust based MTS-DSR protocol improve the routing performance and security in MANET.[13,14]

EXPERIMENTAL SETUP:

The NS-2 which is a discrete event driven simulator developed at UC Berkeley is used in this simulation process. Network Simulator NS-2 is useful in designing new protocols, comparison of different protocols and for evaluating the traffic. A scenario file is taken as input in the NS-2 simulation process that shows the motion of each node and the originating packets by each node. The simulation models are built using the NS-2 version 2.34 and it is run under bandwidth of 40 MHZ.

Table 1: Simulation parameters.

Parameter	Values
Simulation area	600m*600m & 1000m*1000m
Number of nodes	65 & 100
Number of packets sender	25
Constant bit rate	4 (packets/second)
Packet size	512 bytes
Antenna model	Omni directional
Simulation time	500 sec
Malicious Nodes	10 & 20

RESULTS

The results for the above mentioned simulation experiment is shown with the help of graphs. In this section the

comparison of DSR and SDSR is made and the metrics used for the analysis of results are packet delivery ratio and end to end delay, energy.

Table 2: Results of Parameter Values - 1 .

Topology Size = 600M, Number of nodes = 50, Malicious nodes = 10

Packet delivery ratio					
Protocol/NN	10	20	30	40	50
DSR	0.92	0.81	0.72	0.72	0.69
SDSR	0.93	0.82	0.74	0.74	0.70
End – to – End delay					
DSR	0.68	0.62	0.56	0.53	0.51
SDSR	0.45	0.41	0.36	0.29	0.27
Remaining energy					
DSR	0.88	0.85	0.74	0.60	0.57
SDSR	0.78	0.70	0.65	0.57	0.52

Results shows the graph of PDR when the topology size is 600m, the number of malicious nodes 10 and the number of nodes is increased from 10 to 50, Table 2, it is clear that in all acknowledgements based IDS, the suggested scheme SDSR surpasses the performance of DSR which increases PDR by 1.2% when there are 10 to 50 nodes in the network. As the proposed algorithm finds more secure routes with retransmit loss packets frequently, it is possible to increase the delivery ratio. It is observed from and Table 2 that when compared with DSR algorithm, SDSR decreases the delay by 22.4% with the increase in the number of

nodes from 10 to 50. The proposed algorithm SDSR finds the primary and secondary highest forward capacity route in between the sender and receiver. The impact of the number of nodes on the average energy is analysed using the two algorithms and the simulation results are shown in, Table 2 describe the increase in the average energy obtained by the proposed SDSR when there are 10 to 50 nodes. SDSR algorithm reduces the energy by 8.4% as the proposed algorithm to increases no of connections and this system is capable of finding the minimum link failed unbreakable short route between the source and destination.

Table 3: Results of Parameter Values – 2.

Topology Size = 600M, Number of nodes = 100, Malicious nodes = 20

Packet delivery ratio					
Protocol/NN	20	40	40	60	100
DSR	0.88	0.77	0.69	0.68	0.65
SDSR	0.89	0.79	0.70	0.70	0.67
End – to – End delay					
DSR	0.66	0.60	0.54	0.51	0.49
SDSR	0.43	0.39	0.34	0.27	0.25
Remaining Energy					
DSR	0.84	0.81	0.70	0.56	0.53
SDSR	0.74	0.66	0.61	0.53	0.48

Table 3 shows the graph of the PDR when the topology size is 600m, the number of malicious nodes 20 and the number of nodes. The malicious node is varied from 1 to 20 and simulation is carried out to It is clear from the simulation results that the SDSR has the highest delivery ratio in comparison with DSR, when there are 10 to 100 nodes. When the number of nodes increases, the connectivity among the nodes also increases; this enables the proposed method to identify efficient paths which in turn increase the PDR using DSR & SDSR methods. It is observed from Table 3 that when compared with DSR algorithm, SDSR decreases the delay by 22.4% with increase in the number of

malicious nodes from 1 to 20 out of 100 nodes. If find misbehavior node is detected, immediately the SDSR algorithm to use secondary highest forward capacity shortest route in between the sender and receiver. Proposed SDSR increases the remaining energy with the increasing number of malicious nodes from 1 to 20 compared to DSR algorithms. Table 3 describes the increase in the average energy obtained by the proposed SDSR when there are 10 to 100 nodes. SDSR algorithm reduces the energy by 8.4% as the proposed algorithm to increases no of connections and this system is capable of reduce route failed between source and destination.

Table 4: Results of Parameter Values – 3.

Transmission topology = 1000M, Number of nodes = 50, Malicious nodes = 10

Packet delivery ratio					
Protocol/NN	10	20	30	40	50
DSR	0.85	0.74	0.66	0.65	0.62
SDSR	0.86	0.76	0.67	0.67	0.64
End – to – End delay					
DSR	0.64	0.58	0.52	0.49	0.47
SDSR	0.41	0.37	0.32	0.25	0.23
Remaining Energy					
DSR	0.82	0.79	0.68	0.54	0.51
SDSR	0.72	0.64	0.59	0.51	0.46

It is observed from Table 4 that when compared with DSR algorithm, SDSR

increases the delivery ratio by 1.6% with the increase in the number of nodes from

10 to 50. As the proposed algorithm finds maximum secure and lowest link failed route with minimum retransmit packets frequently, it is possible to increase the delivery ratio. SDSR has the lowest delay in comparison with DSR, when there are 1 meter to 1000 meters of transmission ranges. When the transmission range increases, the connectivity among the nodes also increases, which enables the proposed method to identify more number of alternate secondary paths which in turn reduces the delay. Table 4 describes the decrease in delay obtained by the proposed

SDSR when there are 10 to 50 nodes. SDSR algorithm reduces the delay by 22.4%. Results shows the remaining energy when the topology is size 1km, the number of malicious nodes 10 and the number of nodes increased from 10 to 50. The proposed SDSR increases the remaining energy with the increasing topology size compared to DSR. SDSR the proposed algorithm to increases no of connections and this system is capable of reduce route failed between source and destination.

Table 5: Results of Parameter Values – 4.

Topology Size = 1000M, Number of nodes = 100, Malicious nodes = 20

Packet delivery ratio					
Protocol/NN	20	40	40	60	100
DSR	0.79	0.68	0.60	0.59	0.56
SDSR	0.80	0.70	0.61	0.61	0.58
End – to – End delay					
DSR	0.61	0.56	0.49	0.45	0.42
SDSR	0.39	0.35	0.30	0.21	0.19
Remaining Energy					
DSR	0.76	0.73	0.62	0.48	0.45
SDSR	0.66	0.58	0.53	0.45	0.40

Simulation results shows PDR when the topology size is 1000m, the number of malicious nodes 20 and the number of nodes increased from 10 to 100. It is observed from Table 5 that when compared with DSR algorithm, SDSR increases the delivery ratio by 1.6% with the increase in the number of malicious nodes. The impact of the number of nodes on end-to-end delay is analysed using the two algorithms and the simulation results are shown in Table 5. SDSR has the lowest delay in comparison with DSR, when there are 1 meter to 1000 meters of transmission ranges with 20 malicious nodes out of 100 nodes. As the proposed calculation finds the essential and optional strong most extreme forward limit secure connection bombed course, lessening delay is conceivable. The topology size is

varied from 1km to 1000m and simulation is carried out to calculate the average energy using DSR & SDSR methods. Table 5 picture the achieved simulation results, in all acknowledgment-based IDS, the proposed scheme SDSR surpasses DSR performance by 8.4% when there are 10 and 100 nodes in the network. As the proposed algorithm finds the essential and optional strong most extreme forward limit secure connection bombed course, lessening delay is conceivable. SDSR is able to detect misbehaviors in the presence of receiver collision, limited transmission power, false misbehavior report and partial dropping.

From all the above tables, it is clear that the comparison of the SDSR illustrates that the proposed algorithm outperforms

the DSR by providing the highest packet delivery ratio, the lowest end-to-end delay, and the average energy with the increase in the number of malicious nodes.

CONCLUSION

Malicious nodes and loss attack have always been a major threat to security in MANET. In this technical research paper, a novel IDS scheme named SDSR protocol specially designed for MANET is suggested in comparison with the other popular techniques through simulations. The results demonstrate the positive performance of the packet loss and quality of network SDSR when compared with that of DSR in the cases of receiver collision, limited transmission power, false misbehavior report and collaborative attacks.

This paper presents the performance of vary the number of nodes from 50 and 100 in a fixed topography of 600m and 1000 meters with different set of malicious nodes in 10 and 20. SDSR Improve secure transmission packet delivery ratio with quality output and reduce delay ratio with maximum average energy compared to the existing DSR routing protocol were implementing via network simulator 2. To increase the merits of this research work, there is a plan to investigate the following issues in our future research.

- The same concept can be applied in satellite to increase delivery ratio and reduce delay in the route and also to save more energy.
- The performance of SDSR can be tested in real time network environment.

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